PITCH DECK

PRESENTED BY :
Dharmit Shah
Ishaan Chand
Akul Agarwal
Devansh Singh

PRESENTED TO :
CodeX

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PHISHGUARD

PROBLEM STATEMENT

THE "PHISH-SLAYER" BROWSER EXTENSION

PROBLEM

PHISHING ATTACKS TODAY ARE FAR MORE SOPHISTICATED THAN OBVIOUS SCAM EMAILS. MODERN PHISHING WEBSITES CLOSELY IMITATE LEGITIMATE BANKING PORTALS, UPI INTERFACES, AND LOGIN PAGES. THEY USE HTTPS, VALID SSL CERTIFICATES, AND PROFESSIONAL DESIGN LAYOUTS. TO MOST USERS, THESE SITES APPEAR COMPLETELY SAFE.

CURRENT SECURITY SOLUTIONS LARGELY DEPEND ON BLACKLISTS. THE LIMITATION IS THAT BLACKLISTS CAN ONLY BLOCK WEBSITES THAT HAVE ALREADY BEEN IDENTIFIED AND REPORTED. NEWLY CREATED PHISHING DOMAINS, ESPECIALLY ZERO DAY ATTACKS, OFTEN BYPASS THESE SYSTEMS BEFORE THEY ARE ADDED TO ANY DATABASE.

USERS USUALLY RELY ON SIMPLE TRUST INDICATORS LIKE THE LOCK ICON OR FAMILIAR BRAND NAMES. ATTACKERS EXPLOIT THIS BEHAVIOR BY COPYING VISUAL ELEMENTS AND MANIPULATING DOMAIN STRUCTURES TO APPEAR AUTHENTIC. THERE IS A NEED FOR A PROACTIVE SYSTEM THAT EVALUATES WEBSITES INSTANTLY, INSTEAD OF REACTING AFTER DAMAGE IS DONE.

OUR SOLUTION

PhishGuard is a lightweight extension using machine learning for real-time website analysis and protection.

• It examines URL structure, including domain length, suspicious top-level domains, and unusual character patterns.

• The extension scans for login forms, urgent language, and patterns resembling well-known brands.

• SSL certificate evaluation checks validity, issuer info, and recently issued certificates common in phishing.

• A Logistic Regression model processes features locally for fast, private detection without external calls.

• Its a 4 step process involving ml model , llm and api call to urlscan.io

• The system generates dynamic risk scores, classifying websites as Safe, Suspicious, or High Risk.

• It offers speed and privacy by performing all processing and analysis on the user’s device.

TECH STACK

- JavaScript and Chrome Extension Manifest v3
- Python and Scikit learn for model training
- Logistic Regression
- Public phishing datasets such as PhishTank, OpenPhish, and UCI

BUSINESS VIABILITY

The business viability of PhishGuard is supported by several key technical and strategic advantages found in the sources:

• Privacy Advantage: Local ML processing ensures user data privacy, a major selling point for modern security software.

• Operational Efficiency: On-device detection eliminates the need for expensive server-side infrastructure and external API costs.

• Competitive Edge: Real-time analysis identifies new phishing sites that static databases often miss entirely.

• User Trust: Transparent risk scores and clear warnings empower users, building confidence and long-term product loyalty.

• Scalable Architecture: The lightweight, browser-based design allows for easy deployment across various user devices and platforms.

THANK YOU